

## 82. Offcanvas

| Class                          | Meaning             |
|--------------------------------|---------------------|
| <code>.offcanvas</code>        | Hidden side panel   |
| <code>.offcanvas-start</code>  | Start/left          |
| <code>.offcanvas-end</code>    | End/right           |
| <code>.offcanvas-top</code>    | Top                 |
| <code>.offcanvas-bottom</code> | Bottom              |
| <code>.offcanvas-header</code> | Header              |
| <code>.offcanvas-title</code>  | Title               |
| <code>.offcanvas-body</code>   | Body                |
| <code>.offcanvas-sm</code>     | Responsive from sm  |
| <code>.offcanvas-md</code>     | Responsive from md  |
| <code>.offcanvas-lg</code>     | Responsive from lg  |
| <code>.offcanvas-xl</code>     | Responsive from xl  |
| <code>.offcanvas-xxl</code>    | Responsive from xxl |

## 83. Carousel

| Class                                    | Meaning          |
|------------------------------------------|------------------|
| <code>.carousel</code>                   | Main container   |
| <code>.carousel-inner</code>             | Slides container |
| <code>.carousel-item</code>              | Slide            |
| <code>.active</code>                     | Current slide    |
| <code>.carousel-caption</code>           | Caption          |
| <code>.carousel-control-prev</code>      | Previous         |
| <code>.carousel-control-prev-icon</code> | Previous icon    |
| <code>.carousel-control-next</code>      | Next             |
| <code>.carousel-control-prev-icon</code> | Previous icon    |
| <code>.carousel-control-next</code>      | Next             |
| <code>.carousel-control-next-icon</code> | Next icon        |
| <code>.carousel-indicators</code>        | Indicators       |
| <code>.carousel-fade</code>              | Fade transition  |

## 84. Progress

| Class/Property                      | Meaning                  |
|-------------------------------------|--------------------------|
| <code>.progress</code>              | Outer progress container |
| <code>.progress-bar</code>          | Actual progress          |
| <code>style="width:25%"</code>      | 25% progress             |
| <code>style="width:50%"</code>      | 50%                      |
| <code>style="width:100%"</code>     | Complete                 |
| <code>.progress-bar-striped</code>  | Striped                  |
| <code>.progress-bar-animated</code> | Animated stripes         |

## 85. Spinner

| Class                           | Meaning         |
|---------------------------------|-----------------|
| <code>.spinner-border</code>    | Circular loader |
| <code>.spinner-border-sm</code> | Small           |
| <code>.spinner-grow</code>      | Growing loader  |
| <code>.spinner-grow-sm</code>   | Small           |
| <code>.text-primary</code>      | Color           |

## 86. Toast

| Class                         | Meaning                    |
|-------------------------------|----------------------------|
| <code>.toast</code>           | Notification box           |
| <code>.toast-container</code> | Multiple toast positioning |
| <code>.toast-header</code>    | Header                     |
| <code>.toast-body</code>      | Content                    |
| <code>.btn-close</code>       | Close                      |

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## 87. Tooltip

| Attribute/Class                         | Meaning          |
|-----------------------------------------|------------------|
| <code>data-bs-toggle="tooltip"</code>   | Tooltip activate |
| <code>data-bs-placement="top"</code>    | Top              |
| <code>data-bs-placement="bottom"</code> | Bottom           |
| <code>data-bs-placement="start"</code>  | Start            |
| <code>data-bs-placement="end"</code>    | End              |

## 88. Popover

| Attribute                               | Meaning          |
|-----------------------------------------|------------------|
| <code>data-bs-toggle="popover"</code>   | Popover activate |
| <code>data-bs-title</code>              | Popover title    |
| <code>data-bs-content</code>            | Popover content  |
| <code>data-bs-placement="top"</code>    | Top              |
| <code>data-bs-placement="bottom"</code> | Bottom           |
| <code>data-bs-placement="start"</code>  | Start            |
| <code>data-bs-placement="end"</code>    | End              |

## 89. Forms

| Class                         | Meaning                        |
|-------------------------------|--------------------------------|
| <code>.form-label</code>      | Label                          |
| <code>.form-control</code>    | Text/email/password/etc. input |
| <code>.form-control-sm</code> | Small input                    |
| <code>.form-control-lg</code> | Large input                    |
| <code>.form-text</code>       | Help text                      |
| <code>.form-select</code>     | Select                         |
| <code>.form-select-sm</code>  | Small select                   |
| <code>.form-select-lg</code>  | Large select                   |
| <code>.form-check</code>      | ↓<br>Checkbox/radio wrapper    |

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|                                |                        |
|--------------------------------|------------------------|
| <code>.form-check</code>       | Checkbox/radio wrapper |
| <code>.form-check-input</code> | Checkbox/radio         |
| <code>.form-check-label</code> | Label                  |
| <code>.form-switch</code>      | Switch                 |
| <code>.form-range</code>       | Range slider           |
| <code>.form-floating</code>    | Floating label         |
| <code>.input-group</code>      | Input group            |
| <code>.input-group-text</code> | Prefix/suffix          |
| <code>.is-valid</code>         | Valid state            |
| <code>.is-invalid</code>       | Invalid state          |
| <code>.valid-feedback</code>   | Valid message          |

|                                |                 |
|--------------------------------|-----------------|
| <code>.valid-feedback</code>   | Valid message   |
| <code>.invalid-feedback</code> | Invalid message |
| <code>.valid-tooltip</code>    | Valid tooltip   |
| <code>.invalid-tooltip</code>  | Invalid tooltip |

## 90. Form Validation

| Class                          | Meaning             |
|--------------------------------|---------------------|
| <code>.is-valid</code>         | Valid input state   |
| <code>.is-invalid</code>       | Invalid input state |
| <code>.valid-feedback</code>   | Success message     |
| <code>.invalid-feedback</code> | Error message       |



|                                |                       |
|--------------------------------|-----------------------|
| <code>.invalid-feedback</code> | Error message         |
| <code>.valid-tooltip</code>    | Success tooltip       |
| <code>.invalid-tooltip</code>  | Error tooltip         |
| <code>.was-validated</code>    | Form validation state |

## 91. Input Group

| Class                          | Meaning                                                                                   |
|--------------------------------|-------------------------------------------------------------------------------------------|
| <code>.input-group</code>      | Input + addon                                                                             |
| <code>.input-group-text</code> | Prefix/suffix                                                                             |
| <code>.input-group-sm</code>   | Small                                                                                     |
| <code>.input-group-lg</code>   |  Large |

## 93. Ratio

| Class                    | Aspect Ratio |
|--------------------------|--------------|
| <code>.ratio-1x1</code>  | 1:1          |
| <code>.ratio-4x3</code>  | 4:3          |
| <code>.ratio-16x9</code> | 16:9         |
| <code>.ratio-21x9</code> | 21:9         |

| Use             | Example |  |
|-----------------|---------|-------------------------------------------------------------------------------------|
| YouTube video   | 16:9    |                                                                                     |
| Square image    | 1:1     |                                                                                     |
| Cinematic video | 21:9    |                                                                                     |

## 94. Fixed/Sticky Helpers

| Class                       | CSS    | Meaning             |
|-----------------------------|--------|---------------------|
| <code>.fixed-top</code>     | fixed  | Top पर fixed        |
| <code>.fixed-bottom</code>  | fixed  | Bottom पर fixed     |
| <code>.sticky-top</code>    | sticky | Scroll करते समय top |
| <code>.sticky-bottom</code> | sticky | Bottom sticky       |

## 95. Interaction Utilities

| Class                          | CSS                              | Meaning              |  |
|--------------------------------|----------------------------------|----------------------|-------------------------------------------------------------------------------------|
| <code>.user-select-all</code>  | <code>user-select:all</code>     | पूरा text select     |                                                                                     |
| <code>.user-select-auto</code> | <code>user-select:auto</code>    | Default              |                                                                                     |
| <code>.user-select-none</code> | <code>user-select:none</code>    | Text selection रोकना |                                                                                     |
| <code>.pe-none</code>          | <code>pointer-events:none</code> | Pointer events बंद   |                                                                                     |
| <code>.pe-auto</code>          | <code>pointer-events:auto</code> | Pointer events चालू  |                                                                                     |

## 96. Vertical Align

| Class                           | CSS                                     |
|---------------------------------|-----------------------------------------|
| <code>.align-baseline</code>    | <code>vertical-align:baseline</code>    |
| <code>.align-top</code>         | <code>vertical-align:top</code>         |
| <code>.align-middle</code>      | <code>vertical-align:middle</code>      |
| <code>.align-bottom</code>      | <code>vertical-align:bottom</code>      |
| <code>.align-text-top</code>    | <code>vertical-align:text-top</code>    |
| <code>.align-text-bottom</code> | <code>vertical-align:text-bottom</code> |

## 97. Float

| Class                        | Meaning          |
|------------------------------|------------------|
| <code>.float-start</code>    | Start side float |
| <code>.float-end</code>      | End side float   |
| <code>.float-none</code>     | Float remove     |
| <code>.float-sm-start</code> | sm+ start        |
| <code>.float-md-end</code>   | md+ end          |
| <code>.float-lg-end</code>   | lg+ end          |
| <code>.float-xl-start</code> | xl+ start        |
| <code>.float-xxl-end</code>  | xxl+ end         |



## 98. Bootstrap CSS Properties → Utilities

| CSS Property            | Bootstrap Utility       | Example                         |
|-------------------------|-------------------------|---------------------------------|
| <code>display</code>    | <code>d-*</code>        | <code>.d-flex</code>            |
| <code>position</code>   | <code>position-*</code> | <code>.position-relative</code> |
| <code>top</code>        | <code>top-*</code>      | <code>.top-0</code>             |
| <code>bottom</code>     | <code>bottom-*</code>   | <code>.bottom-0</code>          |
| <code>left/start</code> | <code>start-*</code>    | <code>.start-0</code>           |
| <code>right/end</code>  | <code>end-*</code>      | <code>.end-0</code>             |
| <code>z-index</code>    | <code>z-*</code>        | <code>.z-3</code>               |
| <code>width</code>      | <code>w-*</code>        | <code>.w-50</code>              |
| <code>height</code>     | <code>h-*</code>        | <code>.h-100</code>             |



|                |              |               |
|----------------|--------------|---------------|
| height         | h-*          | .h-100        |
| max-width      | mw-*         | .mw-100       |
| max-height     | mh-*         | .mh-100       |
| margin         | m-*          | .m-3          |
| padding        | p-*          | .p-3          |
| gap            | gap-*        | .gap-3        |
| row-gap        | row-gap-*    | .row-gap-3    |
| column-gap     | column-gap-* | .column-gap-3 |
| flex           | flex-*       | .flex-fill    |
| flex-direction | flex-*       | .flex-column  |
| flex-grow      | flex-grow-*  | .flex-grow-1  |



|                 |                   |                         |
|-----------------|-------------------|-------------------------|
| flex-grow       | flex-grow-*       | .flex-grow-1            |
| flex-shrink     | flex-shrink-*     | .flex-shrink-0          |
| flex-wrap       | flex-*            | .flex-wrap              |
| justify-content | justify-content-* | .justify-content-center |
| align-items     | align-items-*     | .align-items-center     |
| align-self      | align-self-*      | .align-self-end         |
| align-content   | align-content-*   | .align-content-center   |
| order           | order-*           | .order-1                |
| text-align      | text-*            | .text-center            |
| font-size       | fs-*              | .fs-3                   |
| font-weight     | fw-*              | .fw-bold                |



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|                  |                       |                |
|------------------|-----------------------|----------------|
| font-weight      | fw-*                  | .fw-bold       |
| font-style       | fst-*                 | .fst-italic    |
| line-height      | lh-*                  | .lh-lg         |
| color            | text-*                | .text-primary  |
| background-color | bg-*                  | .bg-primary    |
| opacity          | opacity-*             | .opacity-50    |
| border           | border-*              | .border        |
| border-width     | .border-1 – .border-5 | .border-3      |
| border-color     | border-*              | .border-danger |
| border-radius    | rounded-*             | .rounded-3     |
| box-shadow       | shadow-*              | .shadow-lg     |





|                             |                                  |                                |
|-----------------------------|----------------------------------|--------------------------------|
| <code>box-shadow</code>     | <code>shadow-*</code>            | <code>.shadow-lg</code>        |
| <code>overflow</code>       | <code>overflow-*</code>          | <code>.overflow-hidden</code>  |
| <code>object-fit</code>     | <code>object-fit-*</code>        | <code>.object-fit-cover</code> |
| <code>visibility</code>     | <code>.visible/.invisible</code> | <code>.invisible</code>        |
| <code>float</code>          | <code>float-*</code>             | <code>.float-end</code>        |
| <code>vertical-align</code> | <code>align-*</code>             | <code>.align-middle</code>     |
| <code>user-select</code>    | <code>user-select-*</code>       | <code>.user-select-none</code> |
| <code>pointer-events</code> | <code>pe-*</code>                | <code>.pe-none</code>          |

## 99. Responsive Utility Formula

| Pattern                                 | Example              | Meaning         |
|-----------------------------------------|----------------------|-----------------|
| <code>.property-value</code>            | <code>.p-3</code>    | सभी breakpoints |
| <code>.property-breakpoint-value</code> | <code>.p-md-3</code> | md+             |
| <code>.text-lg-center</code>            | text + lg + center   | lg+ center      |
| <code>.d-xl-none</code>                 | display + xl + none  | xl+ hide        |
| <code>.mt-lg-5</code>                   | margin-top + lg + 5  | lg+ margin-top  |
| <code>.col-md-6</code>                  | column + md + 6      | md+ 6 columns   |

| 100. Bootstrap JavaScript Components |             |                     |
|--------------------------------------|-------------|---------------------|
| Component                            | JavaScript? | काम                 |
| Accordion                            | ✓           | Sections open/close |
| Alert                                | ✓           | Dismiss             |
| Button                               | ✓           | Toggle state        |
| Carousel                             | ✓           | Slides              |
| Collapse                             | ✓           | Show/hide           |
| Dropdown                             | ✓           | Menu                |
| Modal                                | ✓           | Popup               |
| Navbar                               | ✓           | Responsive collapse |
| Offcanvas                            | ✓           | Side panel          |
| Popover                              | ✓           | Information popup   |

|           |   |                          |
|-----------|---|--------------------------|
| Popover   | ✓ | Information popup        |
| ScrollSpy | ✓ | Scroll के साथ active nav |
| Tab       | ✓ | Tabs switch              |
| Toast     | ✓ | Notification             |
| Tooltip   | ✓ | Tooltip                  |

Bootstrap के interactive components के लिए JavaScript plugins होते हैं; Dropdown, Popover और Tooltip positioning के लिए Popper भी relevant है।  Bootstrap

## 101. Bootstrap JavaScript Methods

| Component | Method               | क्या करता है |  |
|-----------|----------------------|--------------|-------------------------------------------------------------------------------------|
| Modal     | <code>.show()</code> | Modal open   |                                                                                     |
| Modal     | <code>.hide()</code> | Modal close  |                                                                                     |

|          |                        |              |
|----------|------------------------|--------------|
| Modal    | <code>.hide()</code>   | Modal close  |
| Modal    | <code>.toggle()</code> | Open/close   |
| Collapse | <code>.show()</code>   | Open         |
| Collapse | <code>.hide()</code>   | Close        |
| Collapse | <code>.toggle()</code> | Toggle       |
| Dropdown | <code>.show()</code>   | Open         |
| Dropdown | <code>.hide()</code>   | Close        |
| Dropdown | <code>.toggle()</code> | Toggle       |
| Tab      | <code>.show()</code>   | Tab activate |
| Toast    | <code>.show()</code>   | Show         |
| Toast    | <code>.hide()</code>   | Hide         |



|           |                         |                |
|-----------|-------------------------|----------------|
| Bootstrap |                         |                |
| Toast     | <code>.hide()</code>    | Hide           |
| Offcanvas | <code>.show()</code>    | Open           |
| Offcanvas | <code>.hide()</code>    | Close          |
| Carousel  | <code>.next()</code>    | Next           |
| Carousel  | <code>.prev()</code>    | Previous       |
| Carousel  | <code>.pause()</code>   | Pause          |
| Carousel  | <code>.cycle()</code>   | Cycling start  |
| Carousel  | <code>.to(index)</code> | Specific slide |

## 102. Data Attributes

| Attribute                                | काम                     |
|------------------------------------------|-------------------------|
| <code>data-bs-toggle="modal"</code>      | Modal trigger           |
| <code>data-bs-target="#box"</code>       | Modal target            |
| <code>data-bs-toggle="collapse"</code>   | Collapse trigger        |
| <code>data-bs-toggle="dropdown"</code>   | Dropdown                |
| <code>data-bs-toggle="tab"</code>        | Tab                     |
| <code>data-bs-toggle="pill"</code>       | Pill                    |
| <code>data-bs-toggle="tooltip"</code>    | Tooltip                 |
| <code>data-bs-toggle="popover"</code>    | Popover                 |
| <code>data-bs-toggle="offcanvas"</code>  | Offcanvas               |
| <code>data-bs-dismiss="modal"</code>     | Modal close             |
| <code>data-bs-dismiss="modal"</code>     | Modal close             |
| <code>data-bs-dismiss="alert"</code>     | Alert close             |
| <code>data-bs-dismiss="toast"</code>     | Toast close             |
| <code>data-bs-dismiss="offcanvas"</code> | Offcanvas close         |
| <code>data-bs-slide="next"</code>        | Carousel next           |
| <code>data-bs-slide="prev"</code>        | Carousel previous       |
| <code>data-bs-slide-to="2"</code>        | Specific carousel slide |

## 103. Bootstrap CSS Files

| File                                 | इसमें क्या है?            |
|--------------------------------------|---------------------------|
| <code>bootstrap.css</code>           | Complete Bootstrap CSS    |
| <code>bootstrap.min.css</code>       | Minified complete CSS     |
| <code>bootstrap.rtl.css</code>       | RTL version               |
| <code>bootstrap-grid.css</code>      | Grid + flex utilities     |
| <code>bootstrap-utilities.css</code> | Utilities                 |
| <code>bootstrap-reboot.css</code>    | Reboot/base normalization |
| <code>bootstrap.bundle.js</code>     | JS + Popper               |
| <code>bootstrap.js</code>            | JS without bundled Popper |

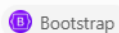


Official Bootstrap distribution में ये अलग CSS/JS builds उपलब्ध हैं।  Bootstrap

## 104. Bootstrap Dark Mode

| Attribute                          | Meaning                                   |
|------------------------------------|-------------------------------------------|
| <code>data-bs-theme="light"</code> | Light color mode                          |
| <code>data-bs-theme="dark"</code>  | Dark color mode                           |
| CSS Variables                      | Theme values को dynamically बदलने में मदद |
| <code>prefers-color-scheme</code>  | System preference detect करने के लिए      |

Bootstrap 5.3 ने color modes और dark mode को CSS variables के साथ expand किया।



## 105. Bootstrap CSS Variables

| Variable        | Meaning       |
|-----------------|---------------|
| --bs-primary    | Primary color |
| --bs-secondary  | Secondary     |
| --bs-success    | Success       |
| --bs-danger     | Danger        |
| --bs-warning    | Warning       |
| --bs-info       | Info          |
| --bs-light      | Light         |
| --bs-dark       | Dark          |
| --bs-body-color | Body text     |

|                      |                         |
|----------------------|-------------------------|
| --bs-body-color      | Body text               |
| --bs-body-bg         | Body background         |
| --bs-border-color    | Border                  |
| --bs-border-radius   | Default radius          |
| --bs-box-shadow      | Default shadow          |
| --bs-font-sans-serif | Default sans-serif font |

Bootstrap 5.2+ में components में CSS custom properties का काफी विस्तार हुआ।



## 106. Bootstrap Sass — Advanced

| Concept                             | क्या करता है                                      |
|-------------------------------------|---------------------------------------------------|
| Sass Variables                      | Bootstrap की default values बदलना                 |
| Sass Maps                           | Colors, spacing, breakpoints आदि की configuration |
| Mixins                              | Reusable Sass logic                               |
| Functions                           | Sass calculations                                 |
| Utility API                         | अपनी custom utility classes generate करना         |
| <code>\$spacer</code>               | Spacing scale का base                             |
| <code>\$grid-breakpoints</code>     | Breakpoints                                       |
| <code>\$container-max-widths</code> | Container widths                                  |
| <code>\$theme-colors</code>         | Theme colors                                      |
| <code>\$enable-dark-mode</code>     | Dark mode control                                 |

|                                   |                   |
|-----------------------------------|-------------------|
| <code>\$enable-dark-mode</code>   | Dark mode control |
| <code>\$enable-shadows</code>     | Shadows           |
| <code>\$enable-rounded</code>     | Rounded corners   |
| <code>\$enable-gradients</code>   | Gradients         |
| <code>\$enable-transitions</code> | Transitions       |

## 107. Bootstrap CSS Grid vs Bootstrap Normal Grid

| Feature     | Normal Bootstrap Grid | CSS Grid              |
|-------------|-----------------------|-----------------------|
| Main system | Flexbox               | CSS Grid              |
| Main class  | <code>.row</code>     | <code>.grid</code>    |
| Child       | <code>.col-*</code>   | <code>.g-col-*</code> |



|                                    |                     |                       |
|------------------------------------|---------------------|-----------------------|
| Child                              | <code>.col-*</code> | <code>.g-col-*</code> |
| Default                            | ✓                   | Separate/opt-in       |
| 12-column concept                  | Yes                 | Configurable          |
| Main spacing                       | Gutters             | <code>gap</code>      |
| <code>grid-template-columns</code> | Main mechanism नहीं | CSS Grid mechanism    |